Modeling and Simulation — Lesson 3
The Mathematics of Epidemics: SIS, SIR, SEIR and other epidemic models

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Introduction to Compartmental Models

Why Model Epidemics?

- Infectious diseases have shaped human history: plague, smallpox, influenza, COVID-19.

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  - Understand **how** and **why** epidemics spread.
  - Predict the course of an outbreak **before** it unfolds.
  - Evaluate **interventions** (vaccination, quarantine, social distancing).
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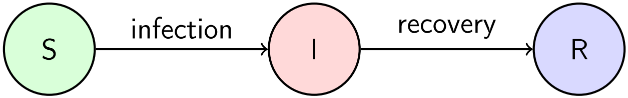
  - Estimate key quantities: peak hospital load, total infections, epidemic duration.
- Models are **simplifications** of reality — but useful ones.
- *"All models are wrong, but some are useful."* — George E. P. Box

Compartmental Models: The Core Idea

- Divide the population into **compartments** based on disease status.
- Individuals move between compartments according to **transition rules**.
- Each transition has a **rate** that depends on the current state of the population.



Example: the SIR model. Susceptible $\rightarrow$ Infected $\rightarrow$ Recovered.

Two Approaches to Epidemic Modeling

Equation-based (ODE) models

- Track **population fractions** (S, I, R).
- Described by differential equations.
- Deterministic — same parameters always give the same result.
- Assume a **well-mixed** population.
- Fast to compute, easy to analyse mathematically.

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Agent-based models (ABM)

- Track **individual agents** with their own state and position.
- Rules applied to each agent separately.
- Stochastic — different runs give different outcomes.
- Can include **space**, **networks**, **heterogeneity**.
- More realistic, but harder to analyse.

Key Concepts We Will Cover

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   - Differential equations, equilibria, stability analysis
   - Basic reproduction number R_0 and herd immunity

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   - Individual agents, random movement, contact-based infection
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2. **Agent-based SIR model**
   - Individual agents, random movement, contact-based infection
   - Stochastic variability and spatial effects
3. **Comparative analysis**
   - When do ODE and ABM agree? When do they differ?
   - Effect of population size, vaccination strategies

A Brief History of Epidemic Modeling

- **1760** — Daniel Bernoulli: first mathematical analysis of smallpox inoculation.
- **1906** — W. H. Hamer: discrete-time model of measles — introduced the **mass action principle** (infection rate $\propto S \times I$).
- **1927** — Kermack & McKendrick: the classical **SIR model** and the epidemic threshold theorem.
- **1950s–70s** — Extensions: SIS, SEIR, age-structured models, vital dynamics.
- **1990s–2000s** — Agent-based and network models become computationally feasible.
- **2020** — COVID-19 pandemic: models (ODE, ABM, network) used worldwide for policy decisions, forecasting, and vaccination planning.

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The mass action principle

The rate of new infections is proportional to the product of susceptible and infected individuals: βSI. This is the foundation of all compartmental models.

Compartment Models

What are Compartmental Models?

- Mathematical approach for describing and analyzing complex systems
- System divided into **discrete compartments** representing specific states or stages
- Transition rates govern the movement of individuals between compartments
- They are often applied to modelling of infectious diseases. The population is assigned to compartments with labels **S**usceptible, **I**nfectious and **R**ecovered).

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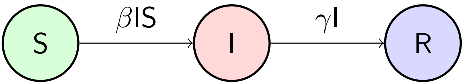
Example of a compartmental model



Design approach to Compartment Epidemiological Models

- **Abstract Models:**
  - *Key question?:* What are general patterns and relationships in epidemics?
  - *Aims:* Investigate general disease behavior. Explore various modes of transmission and countermeasures
  - *Characteristics:* High level overview. Models are simple but not trivial. Direct application to real-world situations may be limited, but it still may be a reasonable starting point.

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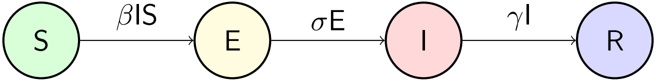
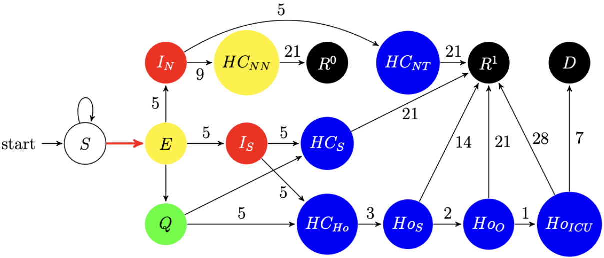
- **Concrete Models:**
  - *Key question?:* What factors are influencing spread of disease in a specific outbreak?
  - *Aims:* Models detailed information about the situation.
  - *Characteristics:* Directly relevant to real-world applications. But lessons learned from one epidemic outbreak may not translate to new situations.

Abstract and concrete model

Basic form of SEIR model



Model used in the Czech Republic at early stages of COVID-19 pandemics



Susceptible-Infected (SI) Model: Overview

- The Susceptible-Infected (SI) model is the simplest form of all disease models. Individuals are born into the simulation with no immunity (susceptible). Once infected and with no treatment, individuals stay infected and infectious throughout their life, and remain in contact with the susceptible population.
- Two compartments: Susceptible (S) and Infected (I)
- Transition rate: infection rate (β)
- Assumes no recovery or immunity
- Model matches the behavior of diseases like cytomegalovirus

SI Model: Equations

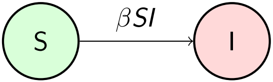
- Susceptible (S): Individuals who are not yet infected but can become infected.
- Infected (I): Individuals who are infected with the disease.

Equations:

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI$$

Diagram:



SI Model: Analytical Solution

Since $S + I = N$ (constant population, normalised to 1), we substitute $S = 1 - I$:

$$\frac{dI}{dt} = \beta(1 - I)I$$

This is the **logistic equation**. Its analytical solution is:

SI Solution

$$I(t) = \frac{I_0}{I_0 + (1 - I_0)\, e^{-\beta t}}$$

- As $t \to \infty$, $I(t) \to 1$ — **everyone gets infected eventually**.
- The growth is initially exponential ($\approx I_0 e^{\beta t}$), then saturates.
- The inflection point (fastest growth) occurs when $I = 1/2$.

Susceptible-Infected-Susceptible (SIS) Model

- The Susceptible-Infected-Susceptible (SIS) model extends the SI model by allowing infected individuals to recover and become susceptible again. Assumes no immunity after recovery, allowing for reinfection
- This model is appropriate for diseases that commonly have repeat infections, for example, the common cold (rhinoviruses) or sexually transmitted diseases like gonorrhea or chlamydia
- Three states:
  - Susceptible (S): Individuals who are not yet infected but can become infected or have recovered from infection and are susceptible again.
  - Infected (I): Individuals who are currently infected with the disease.
- Epidemic occurs the population reaches an equilibrium between susceptible and infective individuals.

SIS Model: Equations and diagram

Transition rates:

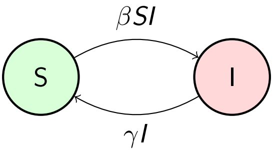
- infection rate (β)
- recovery rate (γ)

Equations:

$$\frac{dS}{dt} = \gamma I - \beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

Diagram:



SIS Model: Stability Analysis

Start from the SIS equation for the infected compartment:

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Expand and collect terms:

$$\frac{dI}{dt} = (\beta - \gamma)\, I - \beta I^2$$

SIS Model: Equilibria and Stability

Equilibria — set $dI/dt = 0$:

$$(\beta - \gamma)\, I^* - \beta\, {I^*}^2 = 0$$

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$$I^* \left[(\beta - \gamma) - \beta I^* \right] = 0 \quad \Rightarrow \quad I_1^* = 0, \qquad I_2^* = 1 - \frac{\gamma}{\beta}$$

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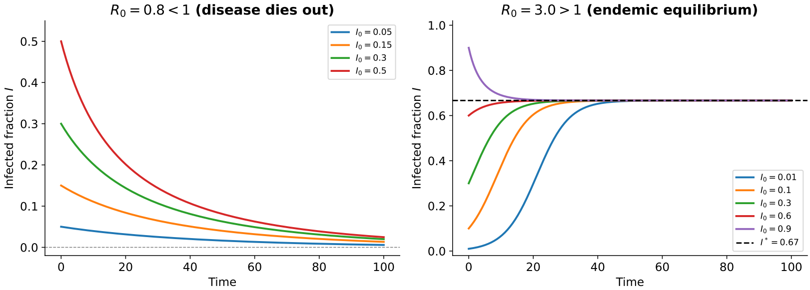
Stability (linearise around each equilibrium):

- **Disease-free** ($I^* = 0$): stable if $\beta < \gamma$, i.e. $R_0 = \beta/\gamma < 1$.
- **Endemic** ($I^* = 1 - \gamma/\beta$): exists and is stable when $R_0 > 1$.

Key insight

In the SIS model, if $R_0 > 1$ the disease persists **forever** at an endemic level $I^* = 1 - 1/R_0$.

SIS Model: Phase Portrait



Left: $R_0 < 1$ — disease dies out. Right: $R_0 > 1$ — endemic equilibrium.

SI Model with vital dynamics

- Extended Susceptible-Infected (SI) model
- Compartmental model in epidemiology considering birth and deaths
- Two compartments: Susceptible (S) and Infected (I)
- Assumes no recovery or immunity
- In addition to the simple model, it captures the impact of population dynamics on disease transmission

Equations and model diagram

Transition Rates

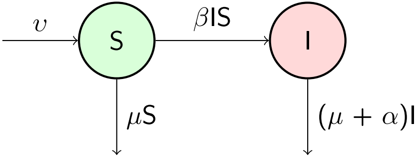
- infection rate (β)
- birth rate (υ)
- datural death rate (μ)
- disease death rate (α)

Equations

$$\frac{dS}{dt} = \upsilon - \beta SI - \mu S$$

$$\frac{dI}{dt} = \beta SI - (\mu + \alpha)I$$

Model Diagram



Susceptible-Infected-Recovered (SIR) Model: Overview

- Widely used compartmental model in epidemiology. Assumes individuals gain immunity after recovery
- Compartments: Susceptible (S), Infected (I), Recovered (R)
- Applications:
  - Studying the spread of diseases where recovery and immunity are possible
  - Evaluating the impact of intervention strategies, such as vaccination or social distancing and predicting the course of an outbreak

SIR Model: Equations and diagram

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- Infection rate (β)
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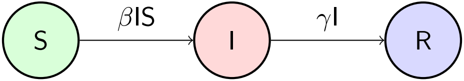

Reproduction Numbers in the SIR Model

Basic Reproduction Number, R_0:

- Expected number of cases generated by one case in a population **where all individuals are susceptible to infection**.
- An epidemic will only occur if $R_0 > 1$.
- R_0 **does not change over the course of an outbreak**, as it is a property of the pathogen and the interaction between the host population and the environment.
- Derived from: $R_0 = \frac{\beta}{\gamma}$ where β is the transmission rate and γ is the recovery rate.

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Effective Reproduction Number, R_t:

- The number of secondary infections. Not all contacts will become infected and the average number of secondary cases per infectious will be lower than R_0.
- R_t is **not fixed** and varies over the course of an epidemic due to interventions and immunizations.
- When R_t falls below 1, the outbreak is under control, leading to a decline in new cases. $R_t = 1$ means endemic.
- The calculation of R_t is complex due to changing dynamics.

Deriving R_0 from the SIR Equations

Consider the early phase when $S \approx 1$ (almost everyone susceptible):

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$$\frac{dI}{dt} = \beta SI - \gamma I \approx (\beta - \gamma)\, I$$

- The infected population grows exponentially if $\beta > \gamma$.
- Define $R_0 = \beta/\gamma$. Then:

$$\frac{dI}{dt} \approx \gamma(R_0 - 1)\, I$$

Epidemic threshold

$$R_0 > 1 \implies \text{epidemic grows} \qquad R_0 < 1 \implies \text{epidemic dies out}$$

Herd Immunity Threshold

The epidemic peak occurs when $dI/dt = 0$:

$$\frac{dI}{dt} = \beta SI - \gamma I = I(\beta S - \gamma) = 0$$

For $I > 0$, this gives $S^* = \gamma/\beta = 1/R_0$.

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Herd immunity threshold

The fraction of the population that must be immune to prevent an epidemic:

$$p_{\text{herd}} = 1 - \frac{1}{R_0}$$

Disease	R_0	p_{herd}
Seasonal influenza	1.3	23%
COVID-19 (original)	2.5	60%
Measles	12–18	92–95%

Final Size of the SIR Epidemic

An important result: the **final size** of the epidemic (total fraction infected) can be found without solving the ODE.
Divide dS by dR:

$$\frac{dS}{dR} = \frac{-\beta S I}{\gamma I} = -R_0\, S \quad \Rightarrow \quad S(t) = S_0\, e^{-R_0\, R(t)}$$

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At the end of the epidemic ($t \to \infty$, $I \to 0$, $R_\infty = 1 - S_\infty$):

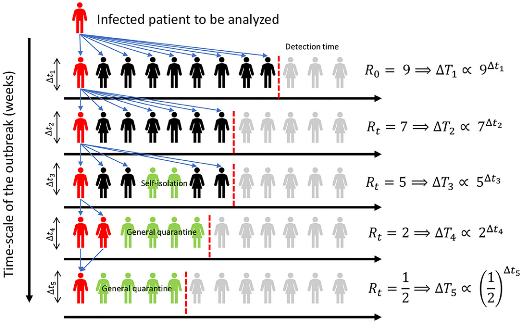
Final size equation

$$S_\infty = S_0\, e^{-R_0\, (1 - S_\infty)}$$

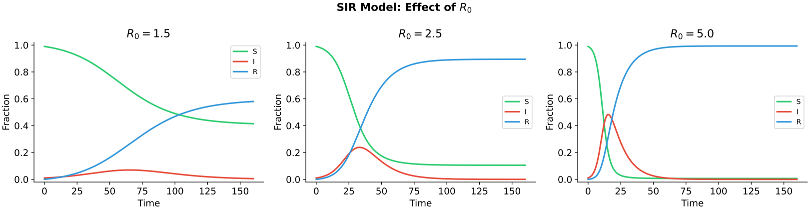
This transcendental equation must be solved numerically, but it tells us that even with $R_0 > 1$, **not everyone gets infected**.

Illustration of reproduction number

- An infected individual (red) can spread the virus among different individuals (black), not reaching part of the population (gray).
- Some individuals go into isolation (green), lowering their contagion chance.
- R_t represents the number of possible new infections caused by a single patient in each outbreak stage.
- In the first days, one individual can infect several people before isolation. As the amount of cases gets public awareness and health policies actions helps to control the outbreak, lowering R_t.

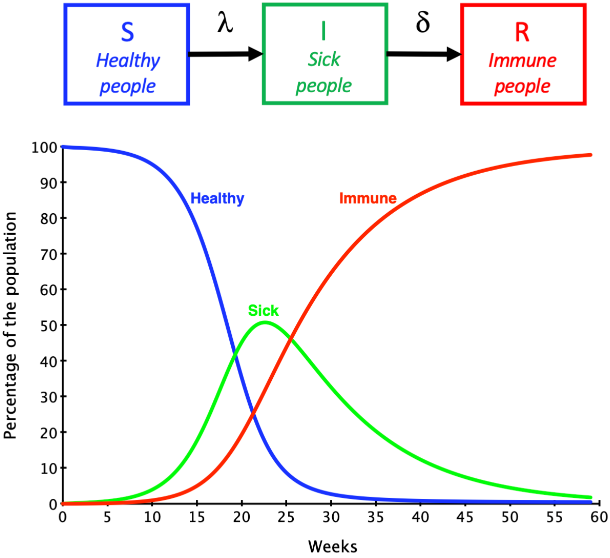


SIR Dynamics: Effect of R_0

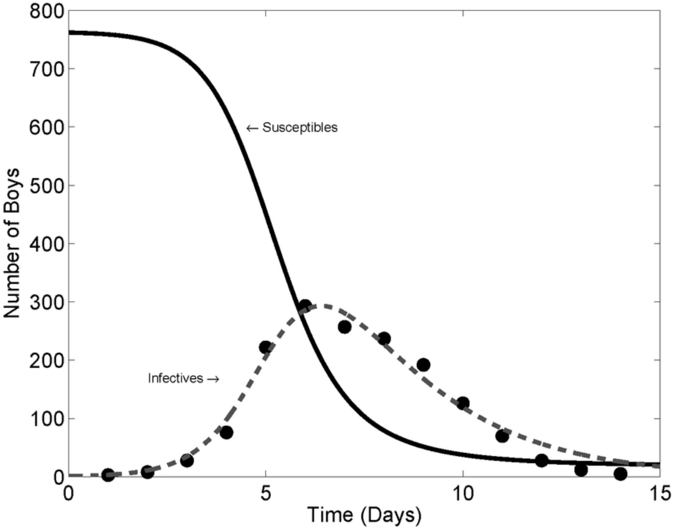


Higher R_0 leads to faster, larger outbreaks with a higher infection peak.

Progress of epidemic by SIR model



Epidemic in English boarding school and SIR model



Susceptible-Exposed-Infected-Recovered (SEIR) Model

- Extended compartmental model in epidemiology. Includes a separate compartment for exposed individuals who are not yet infectious. Suitable for diseases with a latent period where individuals are not infectious immediately after contracting the disease
- Compartments: Susceptible (S), Exposed (E), Infected (I), Recovered (R)
- Exposed individuals (E): have contracted the disease but are not yet infectious; differs from infected individuals (I) who are actively spreading the disease
- Applications:
  - Studying the spread of diseases with latent periods, such as COVID-19, Ebola, or measles
  - Evaluating the impact of intervention strategies, such as vaccination or social distancing and predicting the course of an outbreak

SEIR: Model Equations

Rates

- Infection rate (β): rate at which susceptible individuals become exposed
- Incubation rate (σ): rate at which exposed individuals become infected
- Recovery rate (γ): rate at which infected individuals recover and gain immunity

Equations

$$\frac{dS}{dt} = -\beta SI$$

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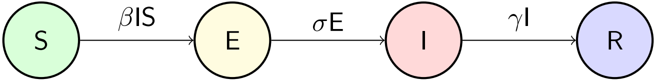
$$\frac{dI}{dt} = \sigma E - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

Diagram



SEIR Model: R_0 and Latent Period

The basic reproduction number for the SEIR model is the same as SIR:

$$R_0 = \frac{\beta}{\gamma}$$

The exposed compartment **delays** but does not change the total number of secondary infections.

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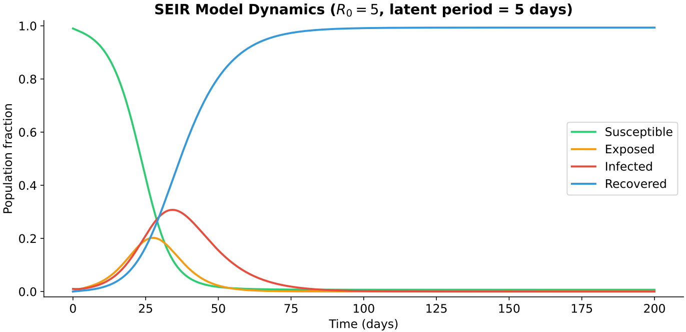
Key parameters:

- **Latent period**: $1/\sigma$ — average time from exposure to becoming infectious.
- **Infectious period**: $1/\gamma$ — average time a person is infectious.
- **Generation time**: $\approx 1/\sigma + 1/\gamma$ — time between successive cases.

SEIR Model: Dynamics



SEIR dynamics with $R_0 = 5$, latent period $= 5$ days. The exposed compartment creates a visible delay between the drop in S and the rise in I.

Applications Beyond Disease Modeling

- Originally developed for epidemiology, but applicable to other fields with similar dynamics
- Key aspects: interacting populations, transitions between states, and time-dependent dynamics
- Examples of alternative applications:
  - Ecology: predator-prey interactions, invasive species spread
  - Social networks: spread of information, opinions, or behavior
  - Computer networks: malware and computer virus propagation

Agent based models

Introduction to Agent-Based Models

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- This approach contrasts with equation-based modeling, where modeled system may be represented via average properties.
- Applications: Epidemiology, social sciences, economics, and more.

Differences between ABMs and Differential Equation Models

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- **Spatial Structure**: ABMs typycally incorporate spatial structures and movement, differential equations models assume well-mixed populations.
- **Complexity**: ABMs handle complex interactions and adaptations, differential equations models focus on average rates of change.

Advantages of Agent-Based Models

- **Emergence**: Emergence occurs when a complex entity has properties or behaviors that its parts do not have on their own, and emerge only when they interact in a wider whole. Think of snow flakes or ant colonies. In ABMs, higher-level system properties may emerge from the interactions of lower-level agents.

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- **Detail**: Ability to model systems at the level of individual agents allows for detailed exploration of wide range of behavior.

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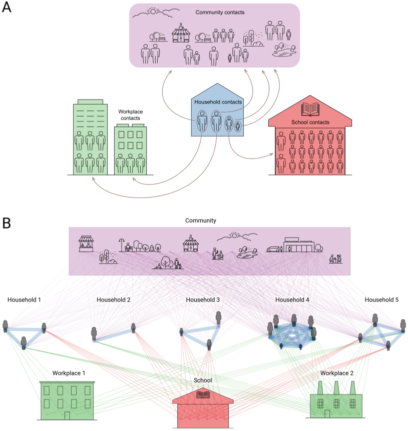
- **Validation Challenges**: Validating the model against real-world outcomes can be complicated due to the complexity of the models.

Applications of Agent-Based Models - Specific Examples

Epidemiology: Modeling the spread of COVID-19 to evaluate the effectiveness of social distancing measures in various communities. Agent-based model *Coves* was developed to help policymakers make decisions based on the best available data, while taking into account the large uncertainties in terms of COVID-19 transmission dynamics, disease progression, and other aspects.



Applications of Agent-Based Models - Specific Examples

Ecology: Simulation of agent-based model of collective nest choice by the ant. Helps in testing the effects and importance of alternative parameter values and decision rules on colonial decision making.

Economics: Analyzing consumer behavior in online stores to predict reactions to changes in price or product availability. May be used for optimizing pricing strategies.

Urban Planning: Planning public transportation systems by simulating the movement patterns of individuals within a city. Helps in making public transport more efficient and accessible.

Social Sciences: Studying the spread of information and misinformation through social networks to understand the dynamics of public opinion formation.

Military: An agent-based computational model for the Battle of Trafalgar. Help to decide what outcomes were possible or what environmental factors has affected the results.

Basics of an Agent-Based SIR Model

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- **Movement**: Each agent performs a random walk (one step per tick) on a bounded grid.
- **Simulation**: Iterative loop updating health states and positions until a set number of steps is reached.

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- **Data Integration**: Can incorporate empirical data at the individual level to calibrate or validate the model.

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When multiple infected agents are nearby, the combined probability is:

$$P = 1 - \prod_i (1 - p_i)$$

Two real-world effects captured

- **Distance matters** — close contacts are far more dangerous than distant ones.
- **Multiple exposures compound** — being surrounded by several infected agents is much riskier.

Simulation Loop in an Agentic SIR Model

- **Update States:**
  - For each susceptible agent, compute the combined infection probability from all infected agents within radius r_{infect}.
  - Use a random draw against this probability to transition from susceptible to infected.
  - For each infected agent, use a random draw with the recovery probability to transition to recovered.
- **Record Results:**
  - Count the number of susceptible, infected, and recovered individuals and store these counts for visualization.
- **Visualization:**
  - Update plots showing current positions and a time-series plot tracking S, I, and R dynamics.
- **Update Positions:**
  - Move each agent using a predefined random movement rule.
  - Ensure that new positions remain within the simulation boundaries.

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- **Parameter exploration**: Varying p_{infect}, p_{recover}, and r_{infect} shows how each parameter shapes the epidemic.

Comparative Analysis & Advanced Topics

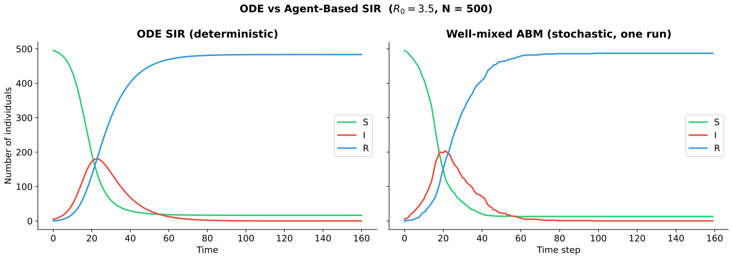
Equation-Based vs. Agent-Based Models

Aspect	Equation-Based	Agent-Based
Unit of analysis	Populations (fractions)	Individual agents
State variables	Continuous (S, I, R)	Discrete per agent
Stochasticity	Deterministic (by default)	Inherently stochastic
Spatial structure	Well-mixed (homogeneous)	Explicit space
Heterogeneity	Hard to include	Natural
Computation	Very fast (ODEs)	Slower (per-agent loops)
Analytical results	Often possible (R_0, stability)	Rarely possible

When to use which?

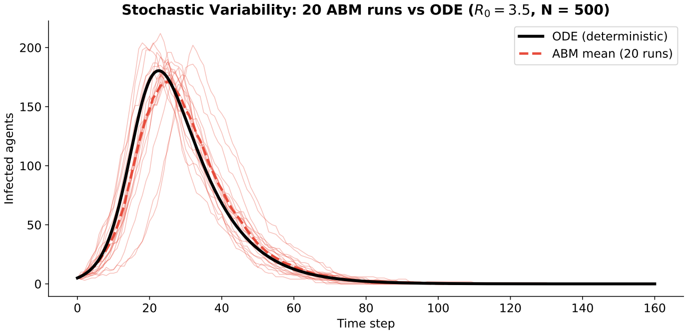
- Use **ODE models** for quick qualitative insight and analytical results.
- Use **ABMs** when individual variation, space, or networks matter.

Comparing SIR Curves: ODE vs. ABM



The ODE model (smooth curves) represents the mean-field limit. The ABM (noisy lines) shows the stochastic reality for finite populations.

Stochastic Variability in the ABM



Multiple ABM runs with identical parameters but different random seeds produce different outbreak sizes. The ODE solution is the deterministic "average" trajectory.

Effect of Infection Radius

The parameter r_{infect} controls how far transmission can reach:

- **Small radius** ($r_{\text{infect}} \approx 1$): Only very close neighbours can transmit — behaves almost like the original same-cell model. Outbreaks spread slowly and may fizzle out.

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- **Large radius** ($r_{\text{infect}} \geq 5$): Each infected agent threatens many neighbours per step, dramatically accelerating spread.
- **Interplay with p_{infect}**: A large radius with low base probability can produce similar dynamics to a small radius with high probability — but the spatial patterns differ.

Practical interpretation

r_{infect} is analogous to the **range of a cough or aerosol cloud**. Social distancing effectively reduces this radius.

Effect of Population Size

- With **small populations**, randomness dominates — outbreaks may fizzle out even when $R_0 > 1$.
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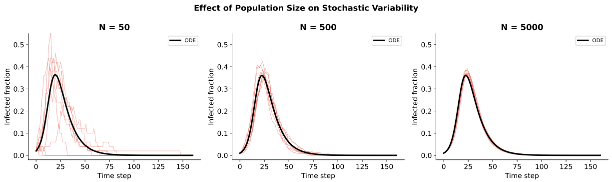
- With **small populations**, randomness dominates — outbreaks may fizzle out even when $R_0 > 1$.
- With **large populations**, the ABM converges toward the ODE solution (**law of large numbers**).
- The probability that an epidemic takes off in a stochastic model is approximately:

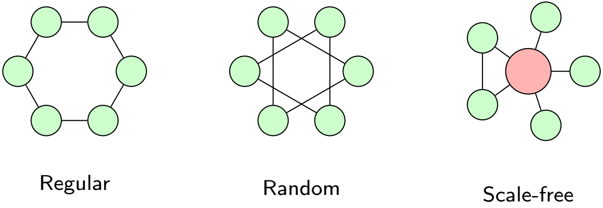
$$P_{\text{outbreak}} \approx 1 - \frac{1}{R_0}$$

for a single initial infected individual.



Network Structure and Disease Spread

- In the ODE model, we assume a **well-mixed** population (everyone contacts everyone).
- In reality, contacts are structured: families, workplaces, schools, transport.
- Network topology strongly affects epidemic dynamics:



Scale-free networks (hubs) make epidemics spread faster and harder to contain.

Vaccination Strategies

- **Random vaccination**: vaccinate a fraction p of the population at random.
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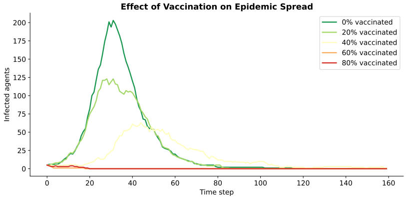
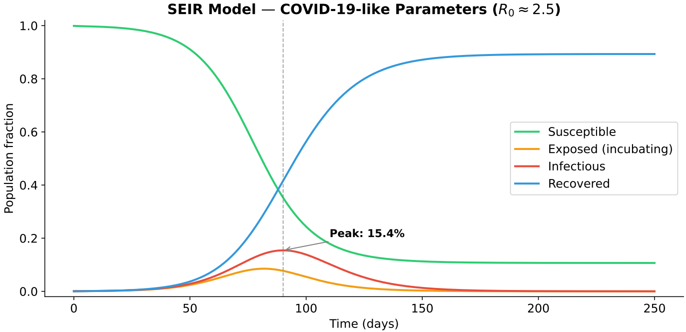
- Epidemic is prevented if $p > p_{\text{herd}} = 1 - 1/R_0$.
- **Targeted vaccination**: vaccinate high-degree nodes (hubs) or high-contact individuals.
- Much more efficient on structured networks.



The SEIR Model in Practice: COVID-19



SEIR model fitted to early COVID-19 data. The exposed compartment captures the incubation period (≈ 5 days) before individuals become infectious.

Summary of Epidemic Models

Model	Compartments	Recovery?	Immunity?	Example disease
SI	2	No	No	Cytomegalovirus
SIS	2	Yes	No	Common cold, gonorrhea
SIR	3	Yes	Yes	Influenza, measles
SEIR	4	Yes	Yes	COVID-19, Ebola

- All models share the same core idea: **compartments** connected by **transition rates**.
- The choice of model depends on the disease biology and the questions being asked.
- R_0 is the single most important parameter across all models.

Exercise: Compare Models

1. Simulate the SIR model with $R_0 = 2.5$ using the equation-based approach. Record the peak infected fraction and the final size.
2. Simulate the same scenario using the agent-based model with 500 agents. Run 10 different seeds.
3. Compare the mean ABM outcome with the ODE result. How close are they?
4. Reduce the population to 50 agents. How does variability change?
5. Explore the effect of r_{infect}: compare $r_{\text{infect}} = 1, 3, 5$ while keeping p_{infect} and p_{recover} fixed. How does the infection radius affect peak timing and outbreak size?
6. Advanced: implement a simple vaccination strategy in the ABM and find the minimum vaccination coverage that prevents an outbreak.